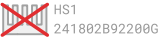


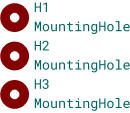
Revision History

Model	Revision	Date	Description
SSDP3A0	A	2025-07	Concept design using the LM70880 buck converter and LM5069-1 hotswap controller. Abandoned due to supply chain issues and cost constraints.
SSDP3A1	A	2025-08	Concept design using the LM5146 buck controller and TPS2490 hotswap controller. Abandoned due to space and cost constraints.
SSDP3A2/SSDV3	A	2025-09	Concept design using the LM5146 buck controller in a custom 1/8th brick form factor. Accepted design for revision.
SSDV3	B	2025-10	Concept design using the cheapest parts available on LCSC. Limited part availability and NRND parts made future production of this revision difficult.
SSDV3	C	2025-10	Prototype design featuring reverse polarity protection in exchange for a smaller TVS diode on the input. Smaller, active-production 3x3mm PQFN-8 FETs used in place of S0-8 FETs. Used for preliminary integration testing and validation.
SSDV3	D	2025-11	Production revision from Rev. C., addressing issues found in testing: - Fixed electrolytic input capacitor unprotected from reverse polarity. - Optimized FETs for Qoss + Qrr losses. - Replaced RPP ground diodes with 100V rated diodes. - Revised capacitor selection for higher temperature rating: X5R -> X7R. - Stricter adherence to voltage clearance requirements. - Refactored schematic for A4 paper. - Removed 1210 capacitors for 1206 to reduce vibration sensitivity.

Heatsink



JLCPCB Eco PCBA Tooling Holes



CAUTION
HOT SURFACE

Mechanical

rpp

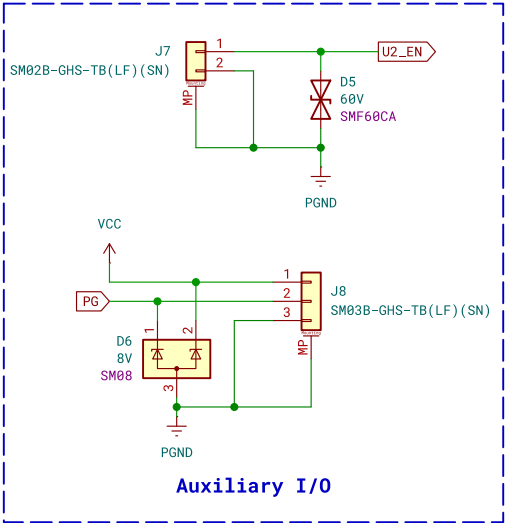


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buck_converter



File: buck_converter.kicad_sch



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Title: Super Step Down V3

Size: A4 Date: 2026-02-06

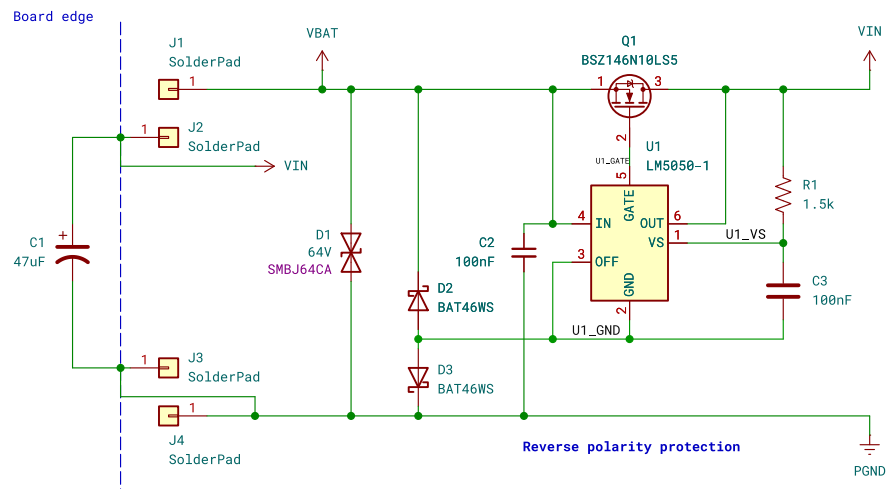
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Rev: D

Id: 1/3

$V_{IN} = 20\sim60V$ (70V slow transient)
typ. 6~12S Li-Ion/LiPo Battery

Attach XT60-M connector via wires to pads



PWR_FLAG PGND
PWR_FLAG VBAT
PWR_FLAG U1_GND

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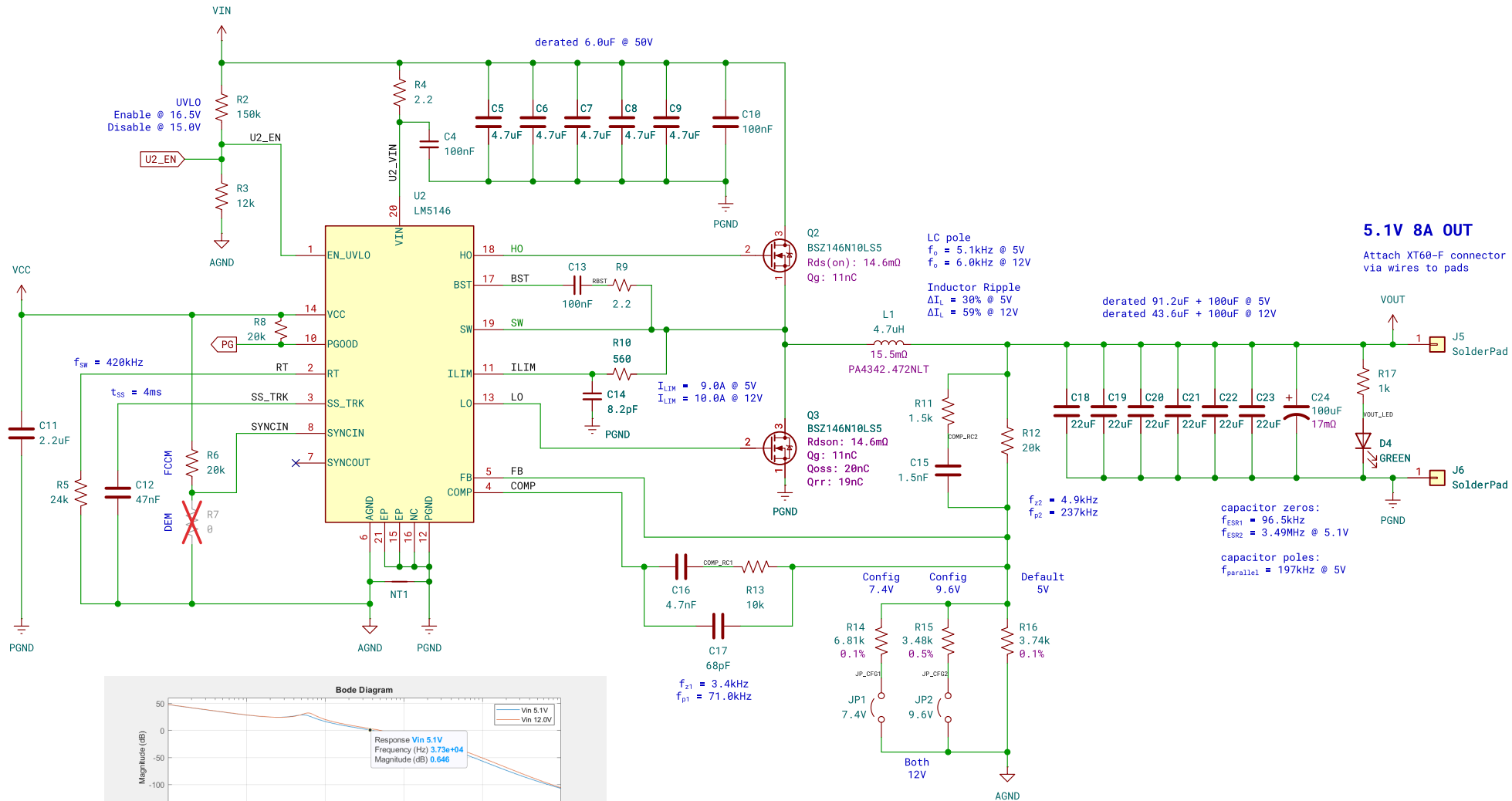
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Rev: D
Id: 3/3